

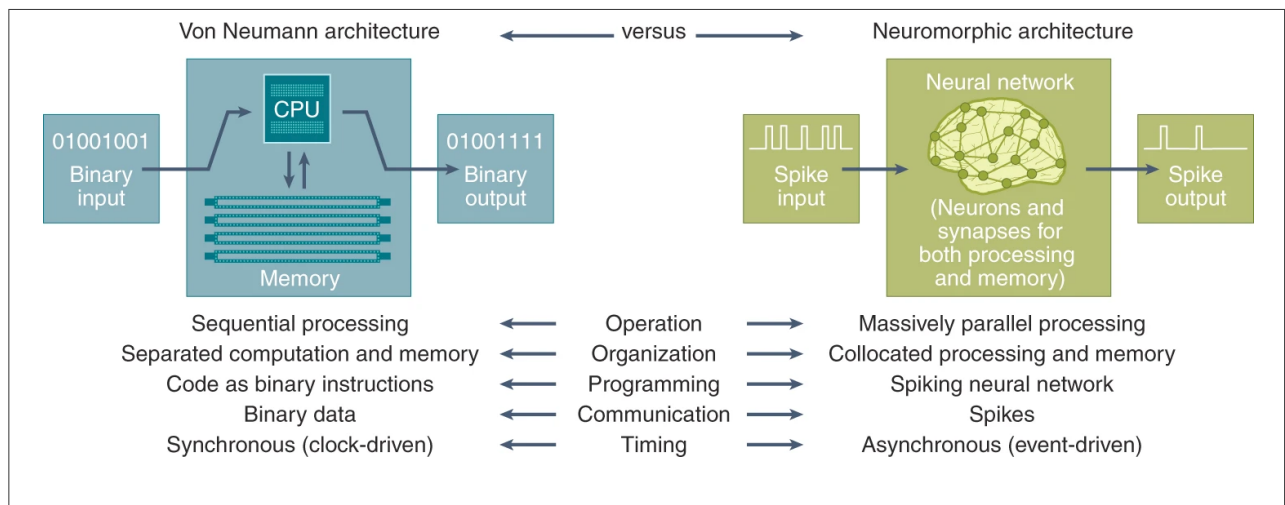


Figure 3: von Neumann vs neuromorphic architectures (Image credits: Schuman, C.D., Kulkarni, S.R., Parsa, M. et al. Opportunities for neuromorphic computing algorithms and applications. Nat Comput Sci 2, 10–19 (2022)).

IBM's TrueNorth

Launched by IBM in collaboration with DARPA's SyNAPSE program, TrueNorth stands as one of the pioneering neuromorphic chips.

Specifications: Comprising 5.4 billion transistors, the chip simulates over a million programmable neurons and 256 million programmable synapses, offering a snapshot of the brain's complexity.

Applications: Beyond mere proof-of-concept, TrueNorth has been deployed in real-world scenarios, including real-time hand gesture recognition, optical character recognition, and even powering a prototype mobile eye-tracking system.

Intel's Loihi

Intel's foray into neuromorphic engineering, Loihi is a testament to the industry's commitment to the paradigm. Designed for research purposes, it has been instrumental in fostering exploration and innovation in neuromorphic applications.

Specifications: The chip simulates roughly 8 million neurons, and is designed to provide high efficiency and scalability.

Potential: With on-chip learning, Loihi represents a significant leap

towards systems that can adapt and evolve. Intel's research has showcased its capabilities in tasks ranging from path planning for robots to adaptive, real-time control systems for prosthetic limbs.

Neuralink

While not a neuromorphic company *per se*, Elon Musk's Neuralink is weaving the narrative of blending advanced computing with neurobiology. Aimed at creating brain-machine interfaces, Neuralink's endeavours underscore the potential of merging the biological with the artificial.

Relevance: Neuralink's work is emblematic of the broader aspiration within neuromorphic computing: understanding, emulating, and eventually synergising with neural systems. As these interfaces evolve, they might incorporate neuromorphic principles to improve efficiency, adaptability, and biocompatibility.

The vision: Beyond medical applications, Neuralink's long-term vision of symbiosis between the human brain and AI systems aligns with the essence of neuromorphic computing – building machines that resonate with the nuances of human cognition.

Why neuromorphic computing is the future of IoT

The future needs devices that not only communicate but also think, adapt, and evolve. Currently, there are two technologies that are powering this vision into reality, i.e., artificial intelligence (AI) and quantum computing.

Role of AI

- **Demand for real-time processing:** AI-driven applications in IoT, from voice assistants to anomaly detection in industrial setups, necessitate real-time data processing. Traditional architectures often grapple with latencies, whereas neuromorphic systems, with their parallelism and event-driven nature, are primed for instantaneous responses.
- **Adaptive learning on the edge:** IoT devices often operate at the edge – away from central servers or clouds. With neuromorphic chips, these devices can learn and adapt on-site without constantly needing data transfers to central servers, ensuring efficiency and data privacy.

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